LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 11. Tuple

Lecture: 121. Tuple Introduction

Section 1: Lecture Summary

This lecture introduces the Python tuple, a built-in data type and data structure that is similar to the list but with a key difference: tuples are immutable. The lecture covers what a tuple is, how to create tuples using various methods, including parentheses, the tuple() function, and multiple value assignment without parentheses. It explains the properties of tuples: they are ordered, allow duplicates, and can contain heterogeneous types of elements. The lecturer contrasts tuples with lists, emphasizing that tuples cannot be modified after creation—no changing elements, adding, or deleting. The immutability makes tuples lightweight and suitable for cases where data is read-only. Finally, the lecture demonstrates how to access tuple elements by index and how to traverse tuples using loops.

Section 2: Key Concepts and Explanations

- What is a Tuple?

  - A tuple is a built-in Python data structure like a list.

  - It is ordered, supports indexing, allows duplicate elements, and can contain heterogeneous types (integers, floats, strings, booleans, complex numbers).

  - Unlike lists, tuples are immutable, meaning their contents cannot be changed once created.

- Differences Between Lists and Tuples

  - Lists use square brackets `[]`; tuples use round brackets `()`.

  - Lists are mutable (elements can be changed, added, or deleted).

- Tuples are immutable (no changes allowed after creation).

- Tuples are lighter weight because they have no resizing or modification overhead.

- Tuples are useful when you want to ensure data is not accidentally modified, especially when data is read-only (e.g., returned by a function).

- Tuple Creation Methods

  - Using parentheses with comma-separated values:

```
t1 = (1, 2, 3, 4, 5)
```

  - Using the `tuple()` constructor with an iterable (list, string, range):

```
t2 = tuple([1, 2, 3])
t3 = tuple('python')
t4 = tuple(range(1, 5))
```

  - Creating an empty tuple:

```
t_empty = ()
```

  - Creating a single-element tuple requires a trailing comma:

```
t_single = (3,)
```

    Without the comma, `(3)` is interpreted as an integer.

  - Tuple packing by assigning multiple values without parentheses:

```
t5 = 10, 11, 12, 13, 14
```

- Tuple Properties and Behavior

  - Ordered: elements have indices starting at 0.

  - Indexing supports positive and negative indices.

  - Immutable: attempts to assign a new value or change elements raise a `TypeError`.

  - Traversal is done using loops similar to lists, e.g. `for x in tuple`.

- Tuple Representation

  - Tuples show elements enclosed in parentheses when printed.

  - Internally, a variable references the tuple object, and the elements exist as separate immutable objects.

Section 3: Example Code and Use Cases

Creating tuples using parentheses

Creating tuples using tuple() function from a list

Creating tuples from a string

Creating tuples from range

Creating an empty tuple

Creating a single-element tuple (note the comma)

Without comma, it's just an integer

Tuple packing (multiple values without parentheses)

Accessing tuple elements by index

Negative indexing

Attempt to modify a tuple (raises error)

Traversing a tuple

```
t1 = (1, 2, 3, 4, 5, 6)
print(t1)  # Output: (1, 2, 3, 4, 5, 6)
print(type(t1))  # <class 'tuple'>

t2 = tuple([1, 2, 3, 4, 5, 6])
print(t2)  # Output: (1, 2, 3, 4, 5, 6)

t3 = tuple('python')
print(t3)  # Output: ('p', 'y', 't', 'h', 'o', 'n')

t4 = tuple(range(1, 5))
print(t4)  # Output: (1, 2, 3, 4)
```

```
t_empty = ()
print(t_empty)  # Output: ()

t_single = (3,)
print(type(t_single))  # <class 'tuple'>

t_not_tuple = (3)
print(type(t_not_tuple))  # <class 'int'>

t5 = 10, 11, 12, 13, 14
print(t5)  # Output: (10, 11, 12, 13, 14)
print(type(t5))  # <class 'tuple'>

print(t1[2])  # Output: 3

print(t1[-1])  # Output: 6

try:
    t1[2] = 10
except TypeError as e:
    print(e)  # Output: 'tuple' object does not support item assignment

for x in t1:
    print(x)
```

Explanation:

- These code snippets demonstrate all principal methods of tuple creation, indexing, the immutability property shown by the error, and iteration over tuple elements.

Section 4: Key Takeaways

- Tuples are immutable sequences in Python, enclosed in parentheses.

- They are similar to lists in that they are ordered, indexed, allow duplicates, and can hold heterogeneous data types.

- Immutability means tuple elements cannot be changed, added, or removed after creation.

- Multiple ways to create tuples include parentheses, the `tuple()` function with an iterable, and implicit packing by multiple assignment without parentheses.

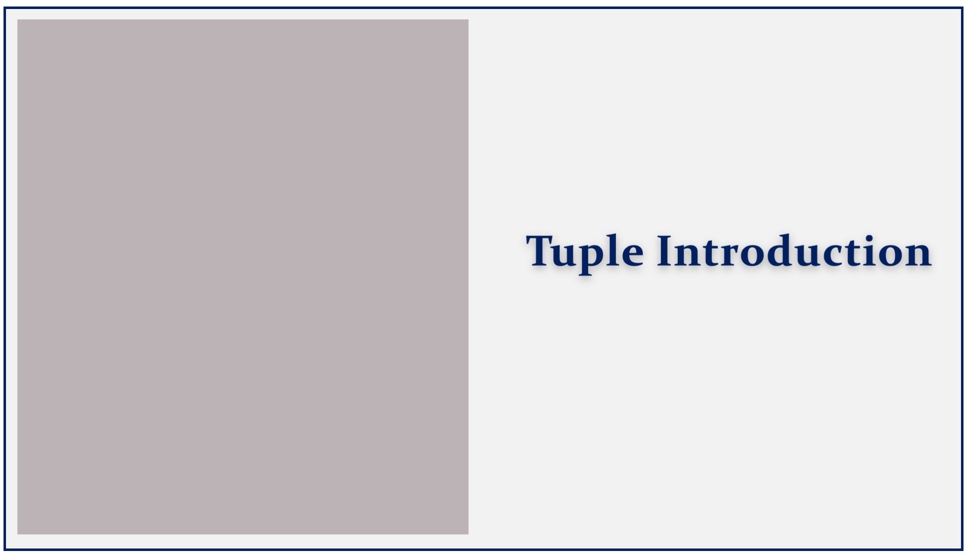
- Single-element tuples require a trailing comma to be recognized as tuples.

- Tuples are lightweight and useful when you want to protect data from modification.

- You can access elements by index and traverse tuples using loops like lists.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Tuple Introduction

Tuple Introduction

What is a Tuple?

Tuple Introduction

What is a Tuple?

Creation

Tuple Introduction

What is a Tuple?

Creation

Representation

Tuple Introduction

What is a Tuple?

Creation

Representation

- immutable

Tuple Introduction

What is a Tuple?

Creation

Representation

- immutable
- Traverse

Tuple Introduction

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

```
t1 = ( 1, 2, 3, 4, 5, 6 )
```

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

```
t1 = ( 1, 2, 3, 4, 5, 6 )
t2 = ( 1.8, 1.8, 3.5, 4.6, 5.7)
```

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

t1 = (1, 2, 3, 4, 5, 6)

t2 = (1.8, 1.8, 3.5, 4.6, 5.7)

t3 = ('john', 'mary' , 'anil')

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

t1 = (1, 2, 3, 4, 5, 6)

t2 = (1.8, 1.8, 3.5, 4.6, 5.7)

t3 = ('john', 'mary' , 'anil')

t4 = (7, 3.2, 'john', True, 5+6j)

Tuple introduction

What is a Tuple ?

Creation

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Tuple Introduction

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Tuple is

Tuple introduction

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t3 = (‘john’, ‘mary’ , ‘anil’)

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Tuple is same as

Tuple introduction

What is a Tuple ?

Creation

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immutable

Traverse

Tuple Introduction

t1 = (1, 2, 3, 4, 5, 6)

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t3 = (‘john’, ‘mary’ , ‘anil’)

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Tuple is same as List

Tuple introduction

What is a Tuple ?

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Tuple Introduction

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t3 = ('john', 'mary' , 'anil')

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Tuple is same as List but immutable

Tuple introduction

What is a Tuple ?

Creation

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Tuple Introduction

Ordered Elements

t1 = (1, 2, 3, 4, 5, 6)

t2 = (1.8, 1.8, 3.5, 4.6, 5.7)

t3 = ('john', 'mary' , 'anil')

t4 = (7, 3.2, 'john', True, 5+6j)

Tuple is same as List but immutable

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

Ordered Elements

t1 = (1, 2, 3, 4, 5, 6)

Duplicate

t2 = (1.8, 1.8, 3.5, 4.6, 5.7)

t3 = ('john', 'mary' , 'anil')

t4 = (7, 3.2, 'john', True, 5+6j)

Tuple is same as List but immutable

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

Ordered Elements

t1 = (1, 2, 3, 4, 5, 6)

Duplicate

t2 = (1.8, 1.8, 3.5, 4.6, 5.7)

t3 = ('john', 'mary' , 'anil')

Heterogenous

t4 = (7, 3.2, 'john', True, 5+6j)

Tuple is same as List but immutable

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

```
t1 = ( 1, 2, 3, 4, 5, 6 )
```

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

```
t1 = ( 1, 2, 3, 4, 5, 6 )
```

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

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Tuple Introduction

```
t1 = ( 1, 2, 3, 4, 5, 6 )
```

```
t2 = tuple ( iterable )
```

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

```
t1 = ( 1, 2, 3, 4, 5, 6 )
```

```
t2 = tuple ( [ 1, 2, 3, 4, 5, 6 ] )
```

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

Tuple Introduction

```
t1 = ( 1, 2, 3, 4, 5, 6 )
```

```
t2 = tuple ( [ 1, 2, 3, 4, 5, 6 ] )
```


```
t3 = ( )
```

Tuple Introduction

Tuple introduction

What is a Tuple ?

Creation

Representation

immutable

Traverse

t1 = (1, 2, 3, 4, 5, 6)

t2 = tuple ([1, 2, 3, 4, 5, 6])

t3 = ( )

t4 = (3)

Tuple Introduction

Tuple introduction

What is a Tuple ?

Creation

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immutable

Traverse

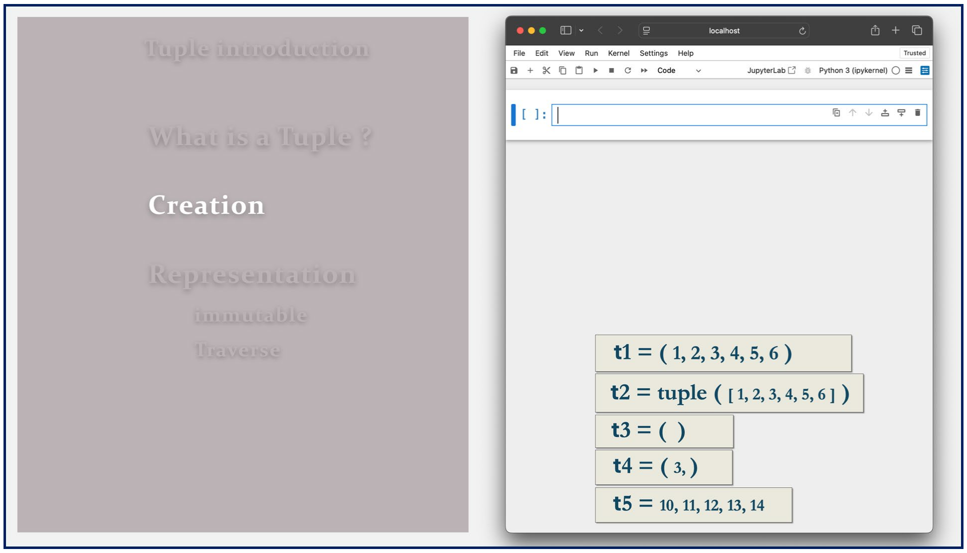
t1 = (1, 2, 3, 4, 5, 6)

t2 = tuple ([1, 2, 3, 4, 5, 6])

t3 = ( )

t4 = (3,)

t5 = 10, 11, 12, 13, 14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Python 3 (ipykernel)
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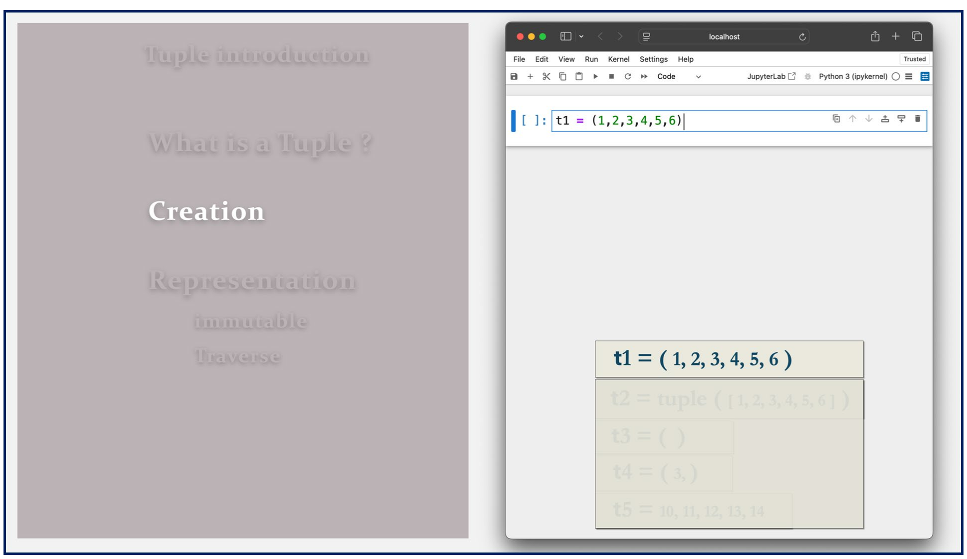
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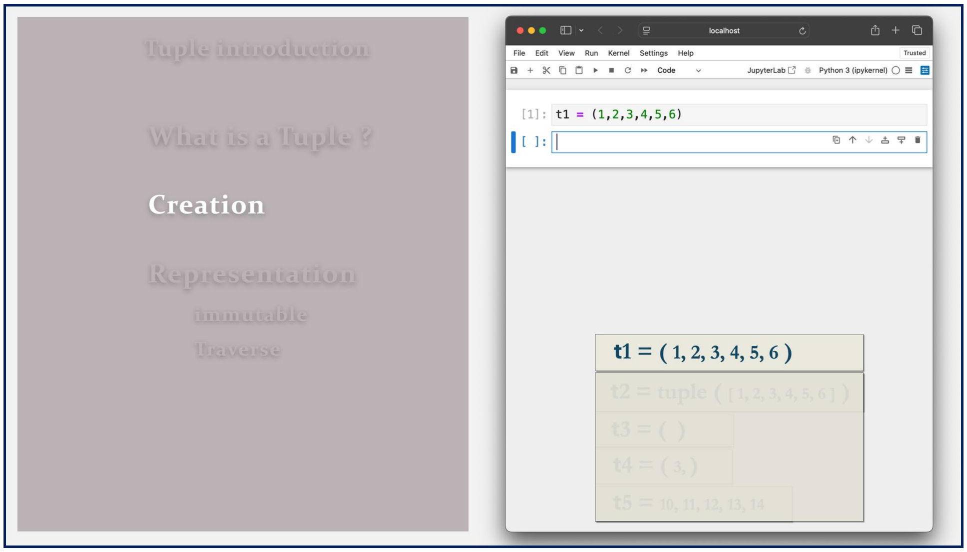
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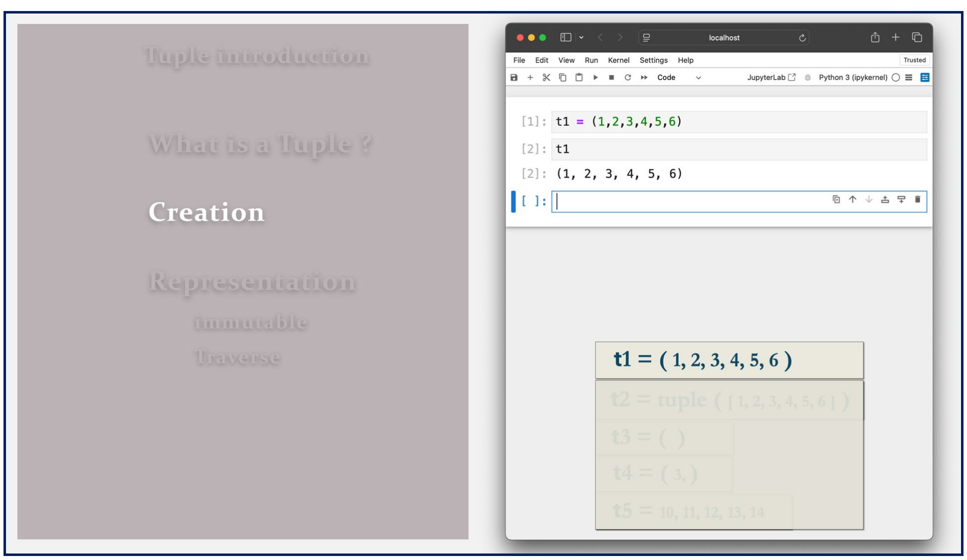
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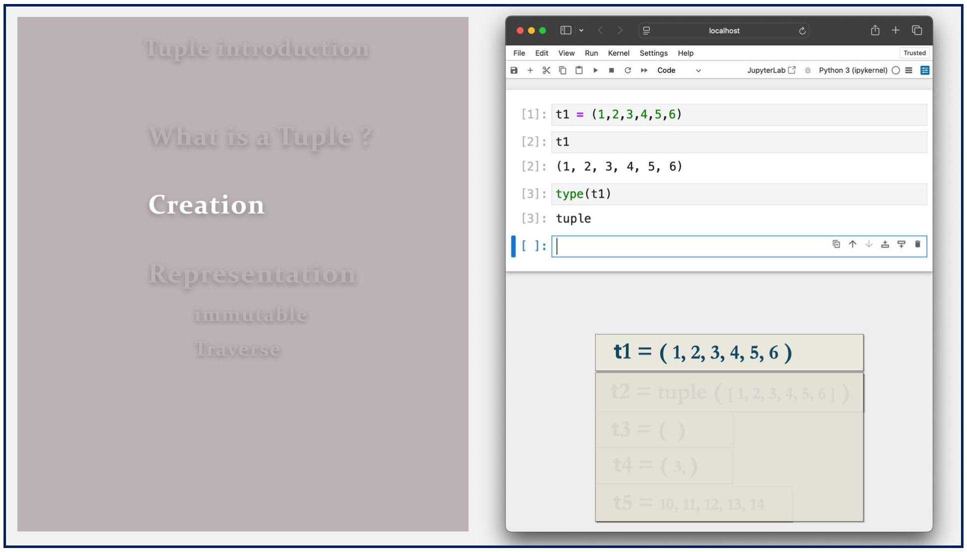
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[2]: t1
[2]: (1, 2, 3, 4, 5, 6)
[3]: type(t1)
[3]: tuple
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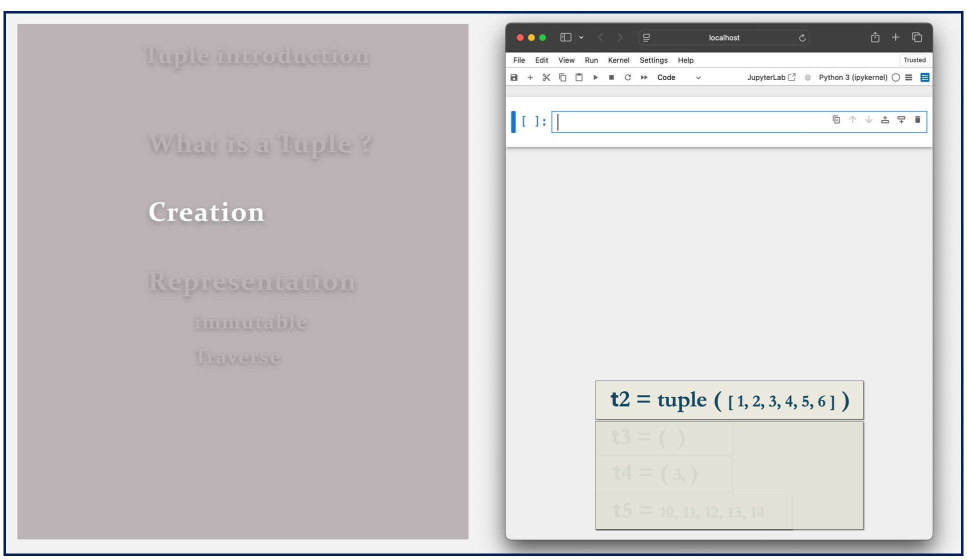
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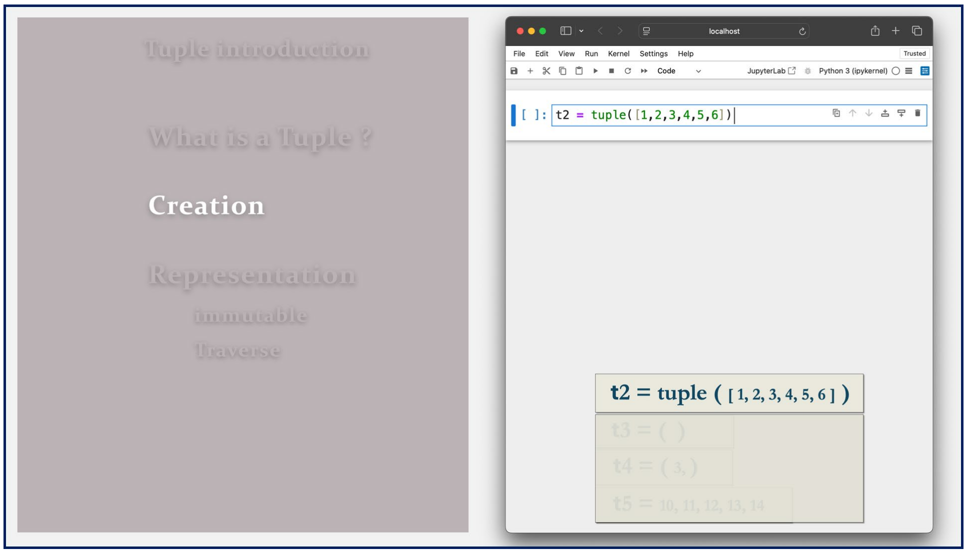

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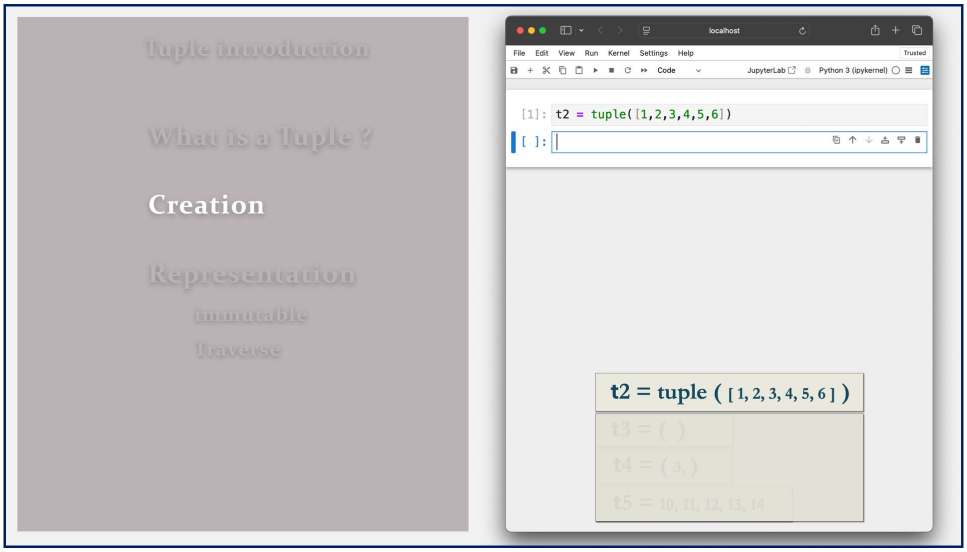

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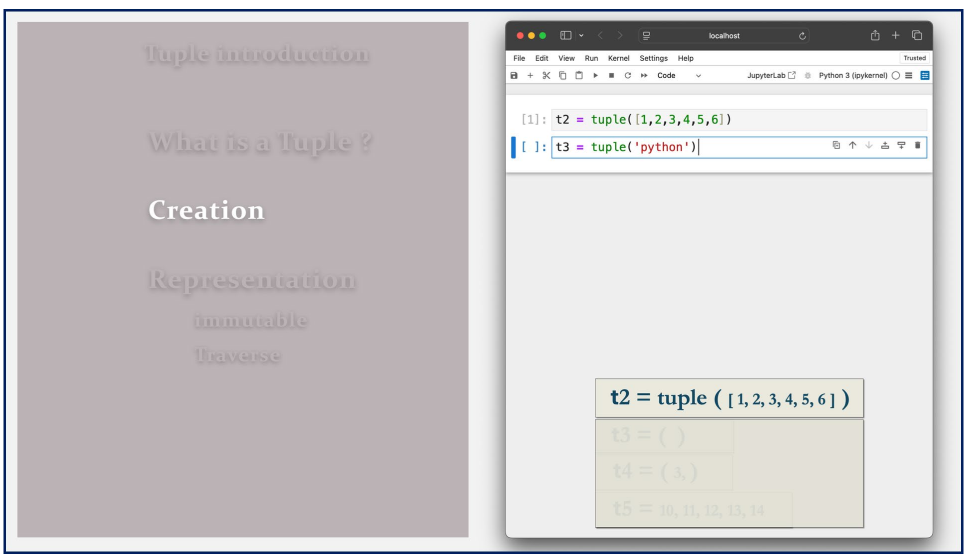

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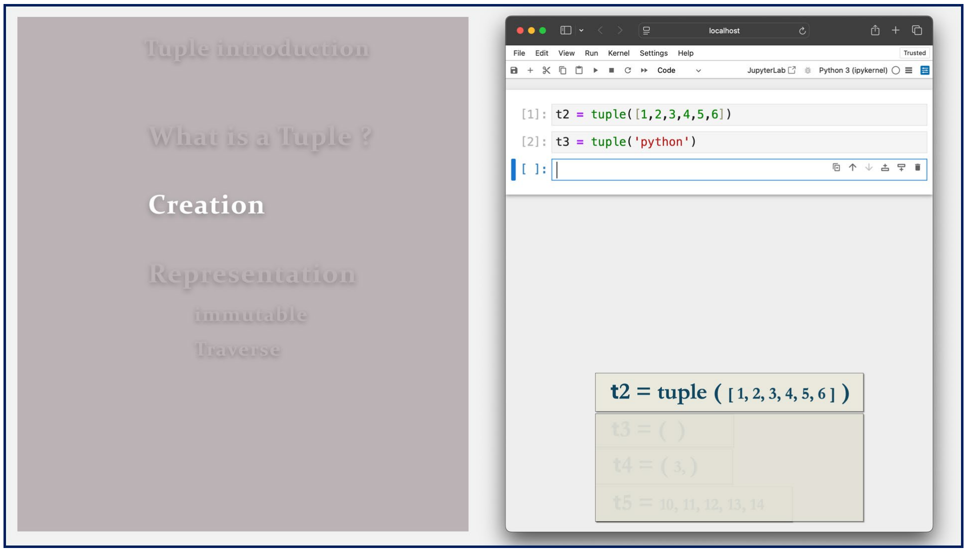
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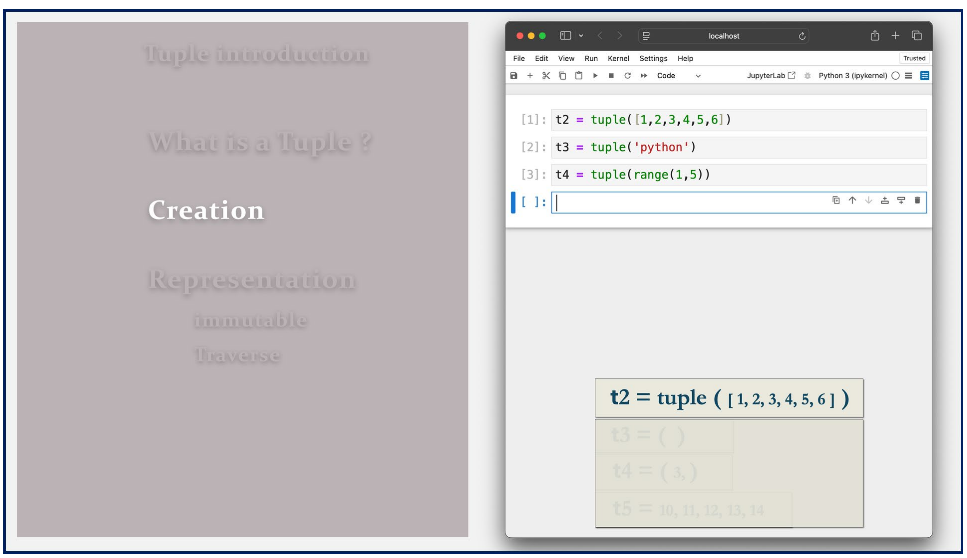
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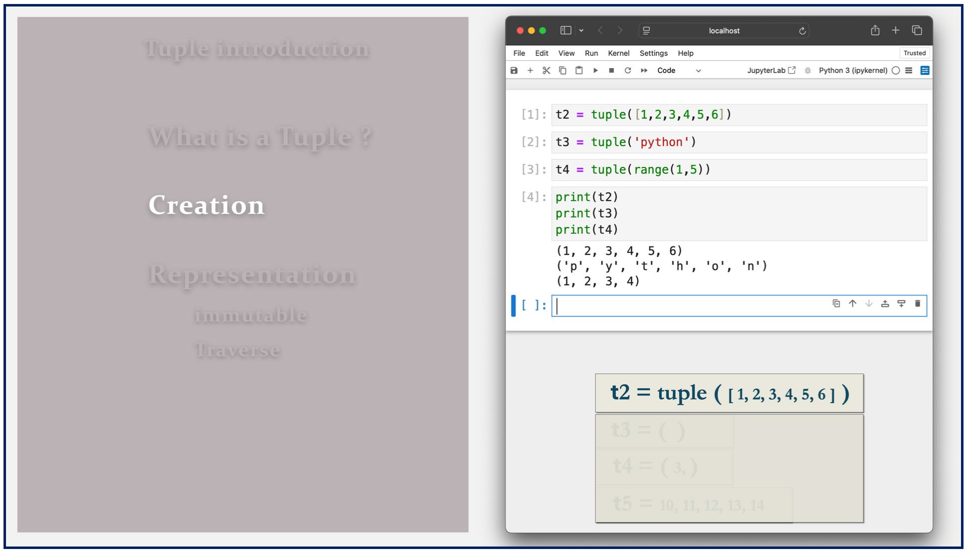
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[2]: t3 = tuple('python')
[3]: t4 = tuple(range(1,5))
[4]: print(t2)
print(t3)
print(t4)
(1, 2, 3, 4, 5, 6)
('p', 'y', 't', 'h', 'o', 'n')
(1, 2, 3, 4)
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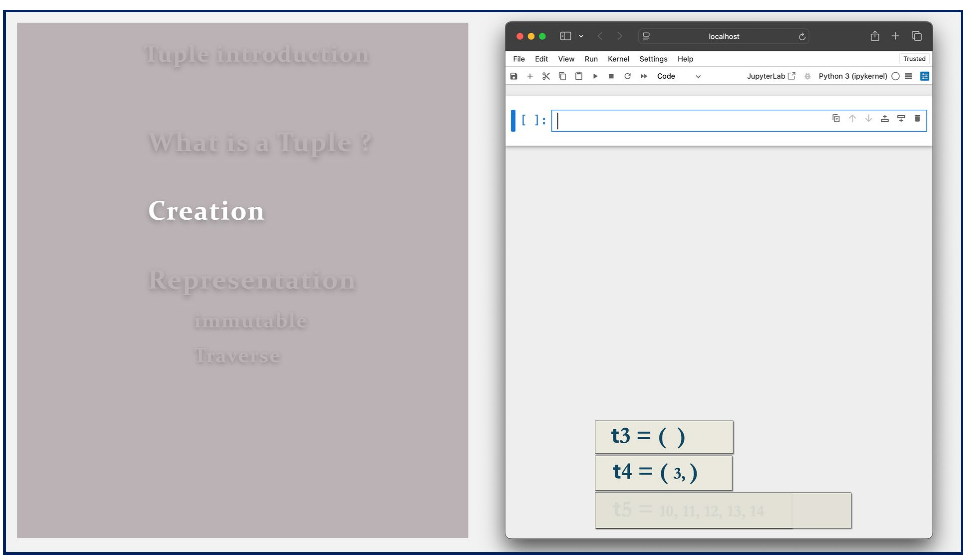
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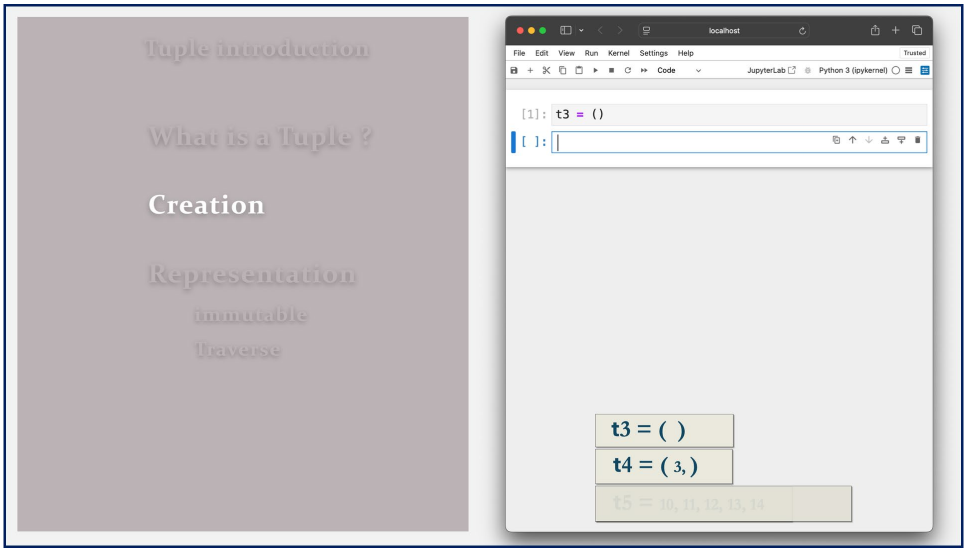
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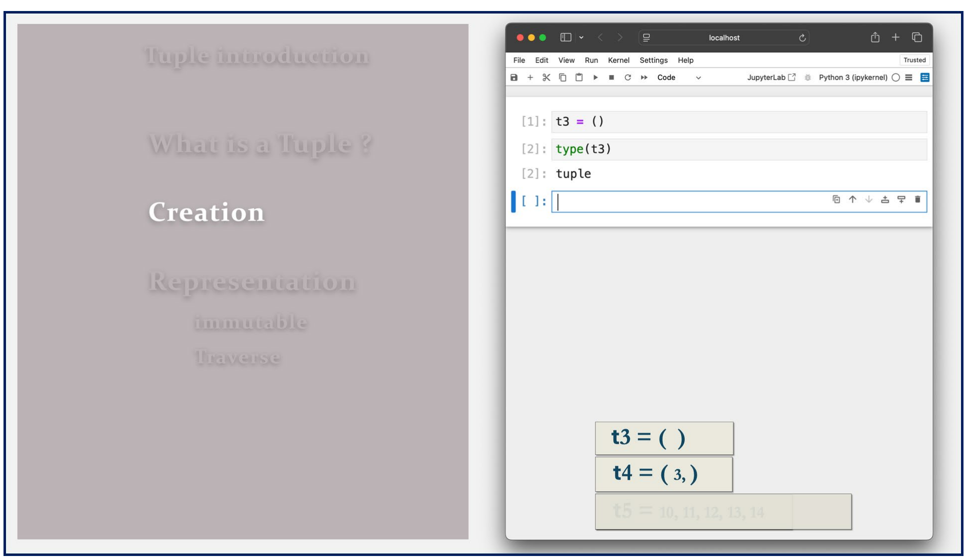
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[2]: type(t3)
[2]: tuple
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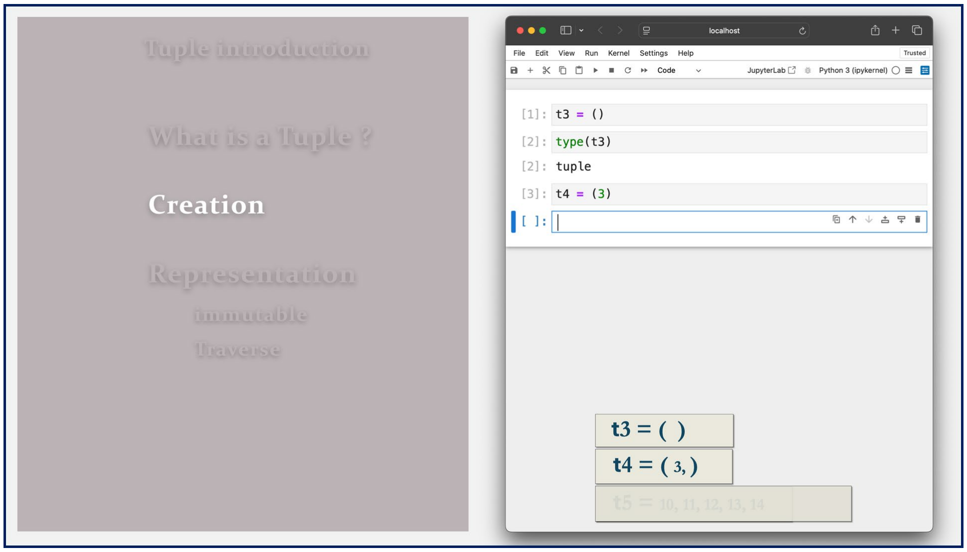
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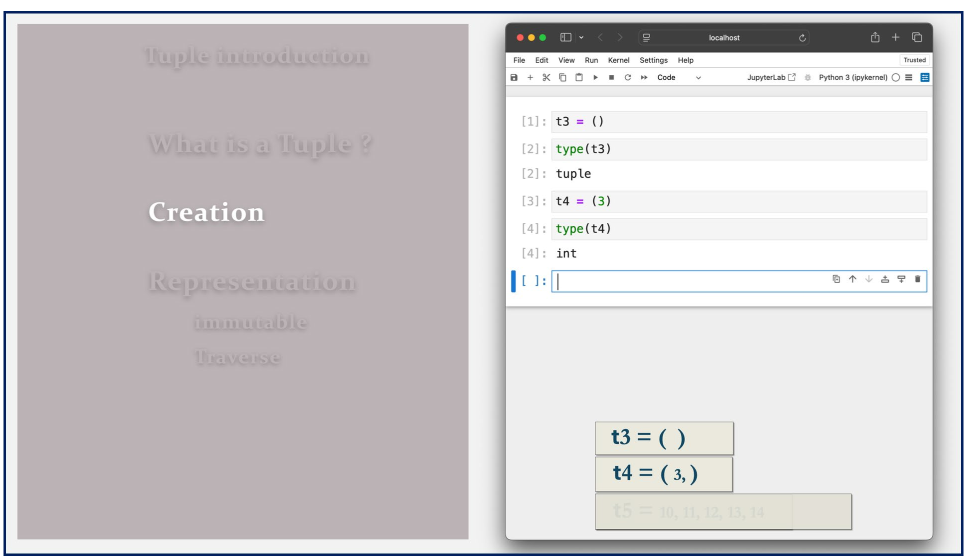
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[1]: t3 = ()
[2]: type(t3)
[2]: tuple
[3]: t4 = (3)
[4]: type(t4)
[4]: int
[ ]:
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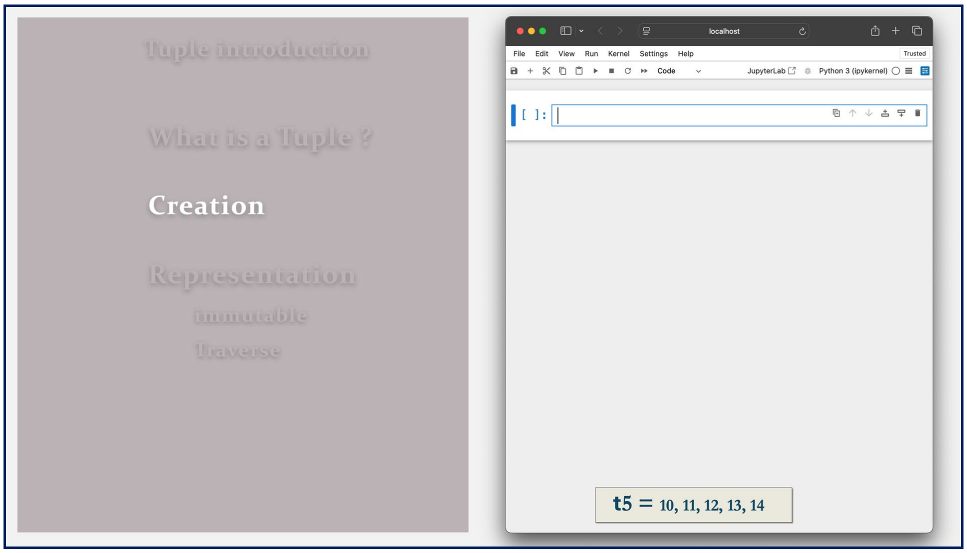
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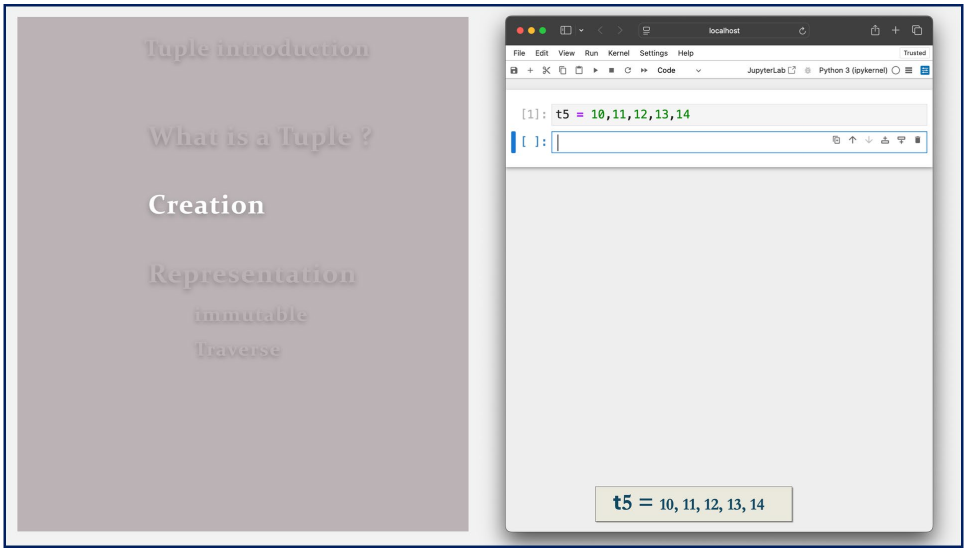
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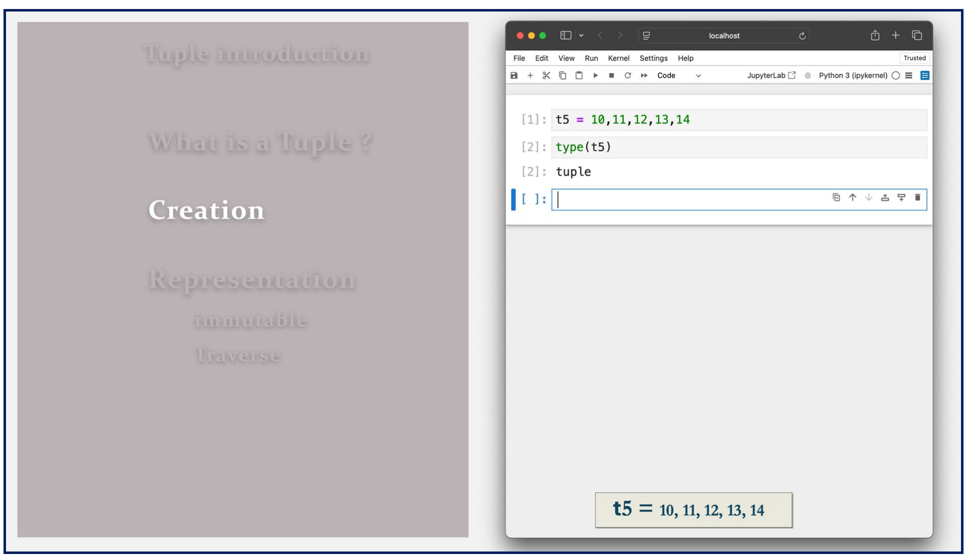
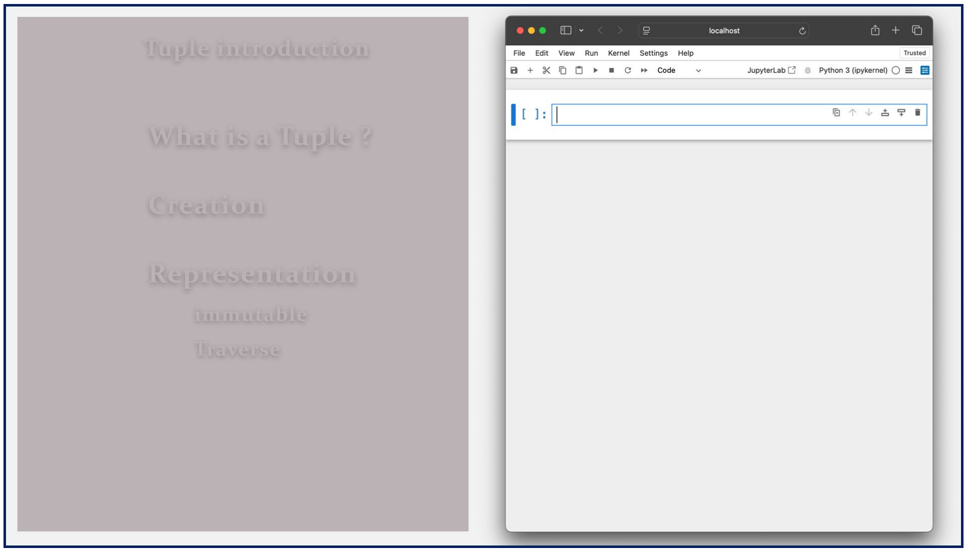
Python 3 (ipykernel)
[1]: t5 = 10,11,12,13,14
[ ]:
t5 = 10, 11, 12, 13, 14


Tuple introduction
What is a Tuple ?
Creation
Representation
immutable
Traverse
localhost
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
[1]: t5 = 10,11,12,13,14
[2]: type(t5)
[2]: tuple
[ ]:
t5 = 10, 11, 12, 13, 14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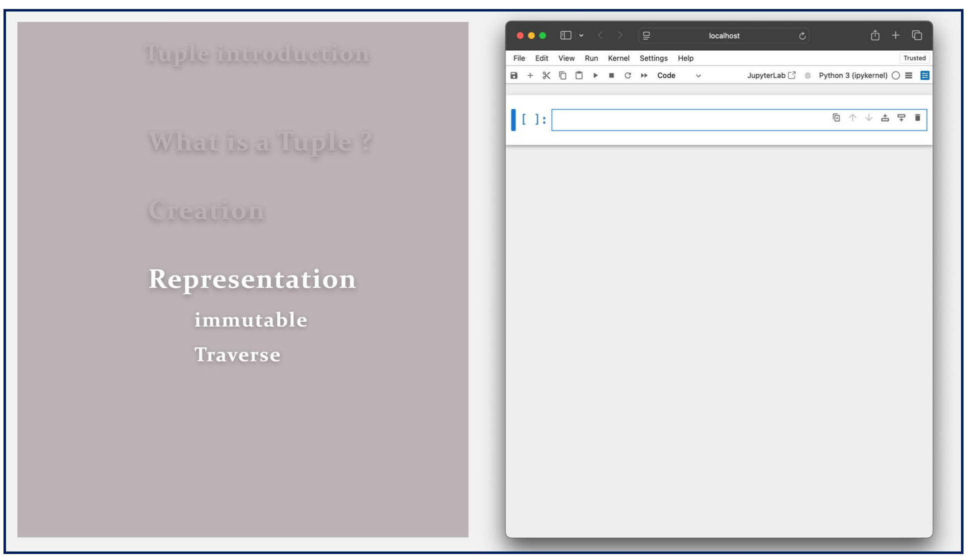
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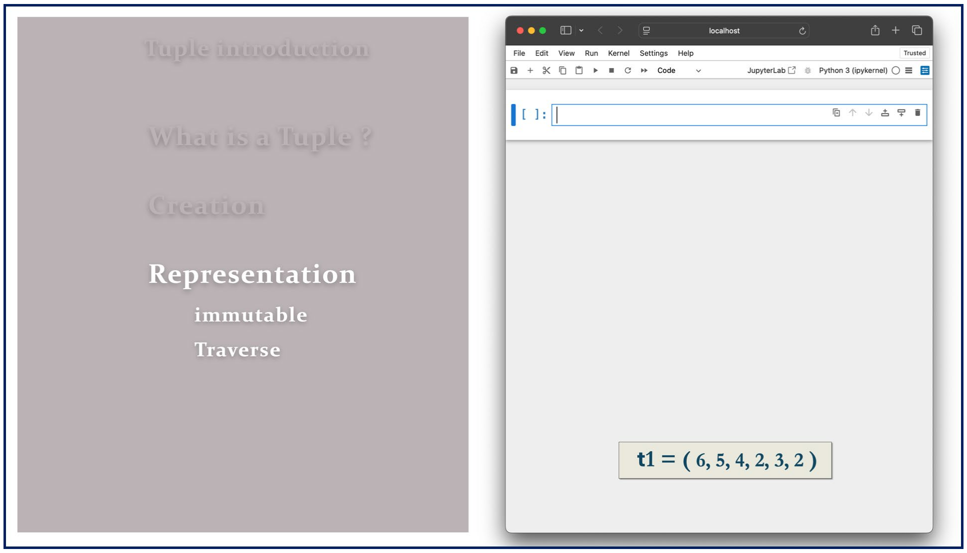
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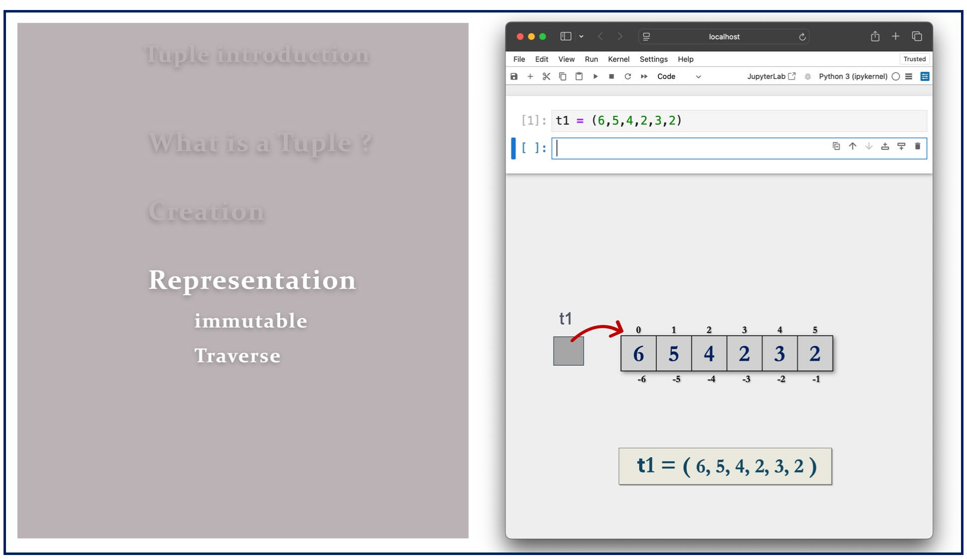
t1 = (6, 5, 4, 2, 3, 2)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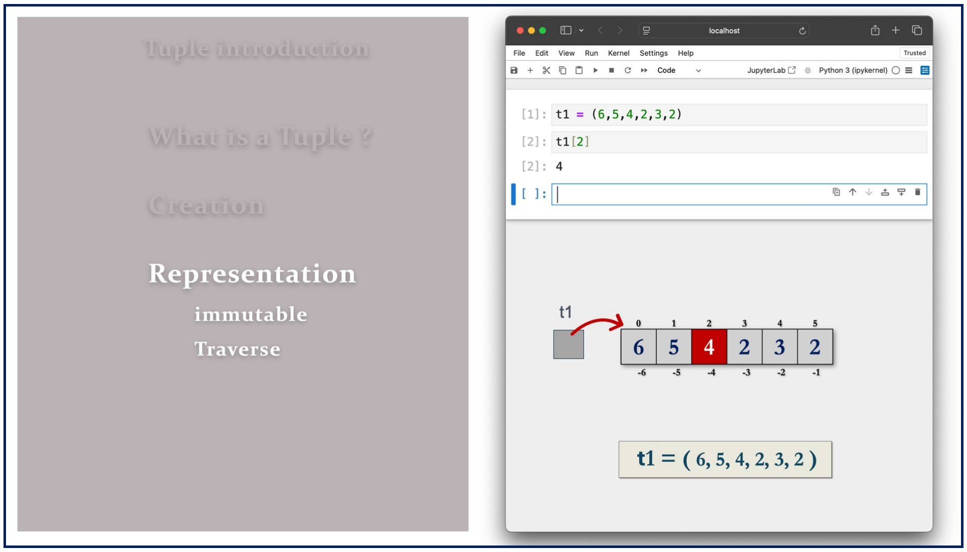
[1]: t1 = (6,5,4,2,3,2)
t1
0 1 2 3 4 5
6 5 4 2 3 2
-6 -5 -4 -3 -2 -1
t1 = (6, 5, 4, 2, 3, 2)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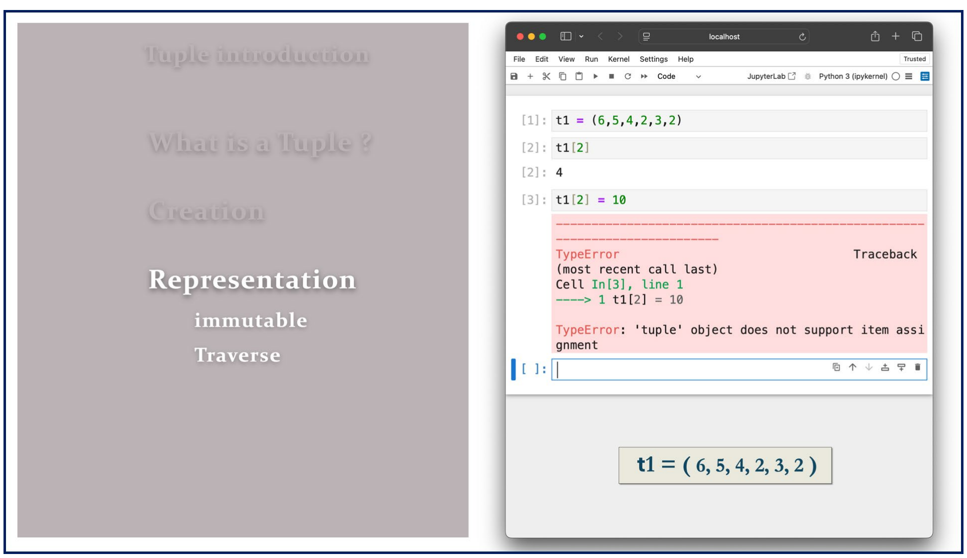
[1]: t1 = (6,5,4,2,3,2)
[2]: t1[2]
[2]: 4
[ ]:
t1
0 1 2 3 4 5
6 5 4 2 3 2
-6 -5 -4 -3 -2 -1
t1 = (6, 5, 4, 2, 3, 2)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Code
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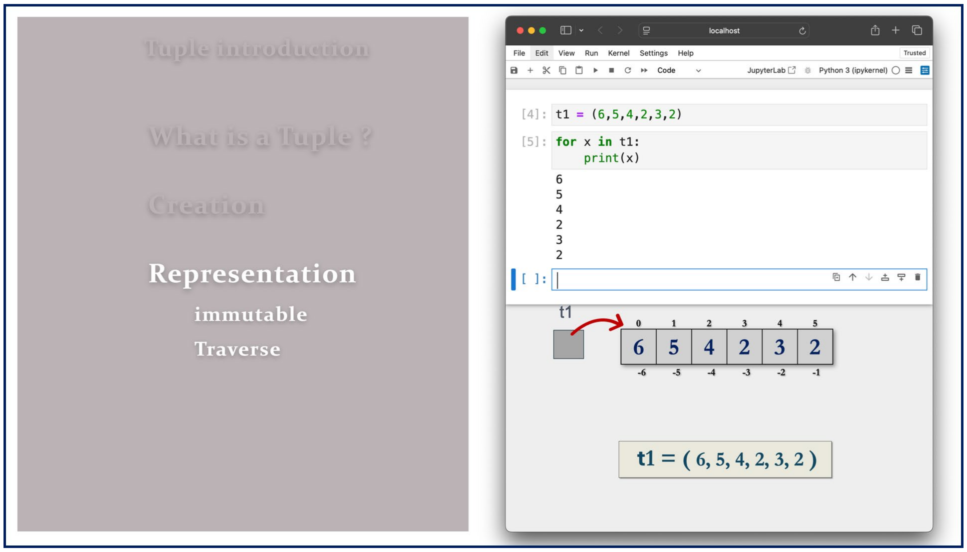
Python 3 (ipykernel)
[1]: t1 = (6,5,4,2,3,2)
[2]: t1[2]
[2]: 4
[3]: t1[2] = 10
TypeError Traceback (most recent call last)
Cell In[3], line 1
----> 1 t1[2] = 10
TypeError: 'tuple' object does not support item assignment
[ ]:
t1 = (6, 5, 4, 2, 3, 2)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Python 3 (ipykernel)
[4]: t1 = (6,5,4,2,3,2)
[5]: for x in t1:
print(x)
6
5
4
2
3
2
[ ]:
t1
0 1 2 3 4 5
6 5 4 2 3 2
-6 -5 -4 -3 -2 -1
t1 = (6, 5, 4, 2, 3, 2)